

Institute of Actuaries of India

Subject ST5 – Finance and Investment A

March 2017 Examination

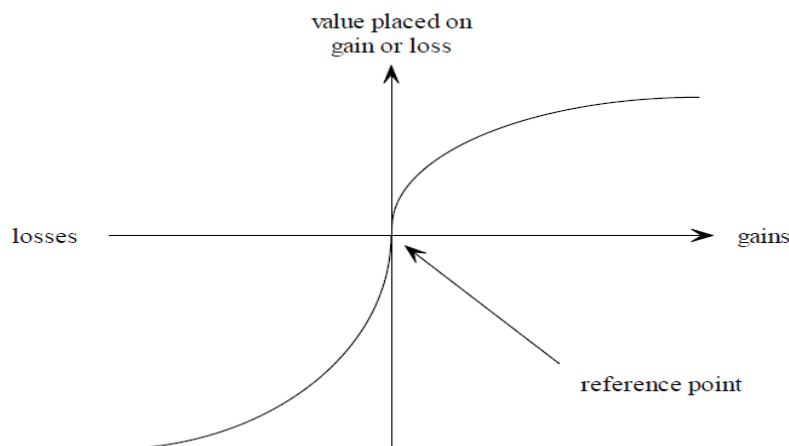
INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- Prospect theory is a theory of how people make decisions when faced with risk and uncertainty.
- It attempts to explain why people may make asymmetric choices when faced with similar possible gains and losses.
- It replaces the conventional risk-averse/risk-seeking decreasing marginal utility theory based on total wealth with a concept of value defined in terms of gains and losses relative to a reference point.
- In particular, individuals are assumed to be risk-averse when facing gains and risk seeking when facing losses defined in terms of some reference point.
- This generates utility curves that are concave above the reference point, convex below the reference point and with a point of inflexion at the chosen reference point.
- Prospect theory therefore suggests that the way in which alternatives are presented or framed can be very important.

**[4 Marks]****Solution 2:****i) Chain-linking**

Adjusting an investment index so that it reflects the investment performance of its constituents (and not injections or withdrawals of funds into or out of the market itself).

The chain-linking process is set up so that:

- the index reflects the new market capitalisation of the constituents
- the index value is not disturbed by the change in capital structure of its constituents.

This is appropriate because if the aim of the index is to reflect the investment performance of its constituents, then the index value should change only in response to that investment performance and not in response to injections or withdrawals of funds into or out of the market itself.

(3)

ii)

- The investor's investment objectives may be specified with reference to one or more indices – eg track or outperform a particular index.
- Indices can be used in the development of the appropriate portfolio to best achieve those investment objectives – by helping us to predict the future possible returns on a particular investment market or sector.
- Indices can be used to value the portfolio and can provide a benchmark against which to monitor its performance – as part of monitoring the investment experience.

(3)

iii)

- a) **Portfolio-based indices** measure rental values, capital values or total returns of actual rented properties. Different indices of this type will give different results because the underlying portfolio of properties will vary in size, regional spread and sector weighting (office, retail etc).

As these actual properties may be sold very infrequently, the valuations upon which the index is based will largely reflect the prices at which comparable properties are sold.

Even these valuations may be out-of-date if the comparables themselves do not come up for sale very often.

The rates of return will typically be money-weighted, meaning that the timing and magnitude of cashflows into the particular property fund will influence the results.

In addition, as the current rental income is fixed until the next rent review, any response to movements in rental values will be sluggish.

- b) **Barometer indices** are designed to act as a barometer of current market conditions.

The barometer type of index aims to track movements in the property market at large by estimating the maximum full rental values of a number of hypothetical rack-rented

properties. As with portfolio-based indices, different barometer indices are produced in practice for different property sectors and (hypothetical) regional spreads.

The values underlying the index are based on valuers' estimates of current rack rents, rather than actual rents, and should therefore give an earlier indicator of changes in market rent levels.

In order to calculate meaningful index values, strict guidelines are required covering both the constituent hypothetical properties to be valued and the assumed lease terms of those properties – eg length of lease, who is responsible for repairs etc.

The main use of this type of index is in highlighting short-term changes in the level of the market in terms of rents and yields.

The main weakness of a barometer index is that the rental values on which it is based are subjective estimates for hypothetical properties – as opposed to actual rents for actual properties.

Portfolio-based indices would be more suitable for this purpose as these can be used for portfolio performance measurement.

With Barometer indices an investor can't closely match its movement with an actual portfolio of property holdings. Hence, these are unsuitable for portfolio performance measurement.

Further, as it is given that majority of flats are rent out, using a barometer index would require subjective estimates for hypothetical properties and hence may not reflect the actual value.

(10)

iv)

The production of reliable indices requires knowledge of the market values of the constituents of the indices at frequent intervals. There are a number of problems in obtaining such information for property:

- Each property is unique and hence almost impossible to have a same property as a reference point. For example, identical flats with same area and layout, would have different view from each balcony.
- The market value of a property is only known for certain when the property changes hands. Just because a property sells for X on a given day does not mean that an identical property next door will also achieve X on an immediate sale.

- Estimation of value is a subjective and expensive process and valuations would have been carried out at different points in time.
- Sales of certain types of investment property are relatively infrequent. For example, very few large shopping malls are bought/sold in any one day!
- The prices agreed between buyers and sellers of properties are normally treated with a degree of confidentiality.
- The heterogeneity of property magnifies the problems of obtaining price data. It is difficult to group properties into usefully homogeneous groups and still obtain sufficient price data for each group.
- The mix of transactions by type may vary period by period, complicating the construction of weights to use in indices.

(6)

[22 Marks]**Solution 3:**

i)

The theoretical price of the future

$$= 100 \times \exp \{(0.08 - 0.03) \times 6/12\}$$

$$= 102.53$$

The actual future price is 105. This shows that the index futures price is too high relative to the index.

The correct arbitrage strategy would be to:

- i. Sell future contracts
- ii. Buy the shares underlying the index

(3)

ii)

Storage cost of INR 12 per year payable monthly means INR 1 per month.

The present value of the storage costs for four months are

$$1 + 1 \times \exp \{(-0.08) \times 1/12\} + 1 \times \exp \{(-0.08) \times 2/12\} + 1 \times \exp \{(-0.08) \times 3/12\}$$

$$= 3.960$$

The future price would be

$$= (2500 + 3.960) \times \exp \{ (0.08) \times 4/12 \}$$

$$= 2571.63$$

(3)

iii)

The pros of investing e-gold are:

- + no hassles of storage and safe keeping
- + no concerns of physical theft
- + easy to keep a track of quantity of gold held by an individual or family and once it passes from one generation to another
- + easy to declare and pay taxes as and when applicable
- + one doesn't have to be concerned about the purity of the gold and likely fraud and cheating cases of not getting the 99.99% purity rate at the time of sale
- easy to break and sell a portion of the total investment, e.g. if one has 100 gm of e-gold, then one could potentially sell 1 gm of gold, if the tradeable on the exchange, whereas it may be difficult to sell 1% of jewelry, unless held in the form of gold bullion

The cons are:

- The sentimental value of physical possession of gold and ornaments is not present
- difficult to gift, receive and share with friends and family
- one really cannot wear an e-gold during functions and ceremonies and hence goes against the fashion or style statements
- due to a likely lock-in, it may not be feasible to sell in times of immediate requirement

(5)

iv)

Assuming INR 12 to be spread evenly over the year, hence taking midway during the year.

The total storage cost for three years will be:

$$12 \times [\exp \{ (-0.08) \times 1/2 \} + \exp \{ (-0.08) \times 3/2 \} + \exp \{ (-0.08) \times 5/2 \}] = 32.00$$

Hence, a discount of INR 32, would make the individual neutral between the two modes.

(2)

[13 Marks]

Solution 4:

i)

Monetary policy – the control of some measures of the money supply and/or the level and structure of interest rates. This is a particularly effective tool in controlling the rate of inflation.

Fiscal policy – decisions on the level and structure of taxation and government expenditure and hence, the public sector borrowing requirement (or debt repayment). The public sector borrowing requirement (PSBR) or fiscal deficit is the excess of government expenditure over government income from taxation. If this quantity is negative it is known as the public sector debt repayment (PSDR) of fiscal surplus.

National debt management policy – the manipulation of the outstanding stock of government debt instruments held by the domestic private sector, in order to influence the level and structure of interest rates or the availability of liquid reserve assets to the banking sector. This policy is closely inter-linked with monetary and fiscal policy.

Exchange rate policy – directed towards achieving some target for the exchange rate of the domestic currency in terms of foreign currencies. The lower the value of the domestic currency and the local currency cost of factor inputs (land, labour and capital), the more competitively priced will be the goods and services produced.

Prices and incomes policy – aimed at influencing the rates of wage and price inflation. Prices and incomes policies aim to control these variables (and hence inflation) directly by imposing maximum increases. Governments have encountered practical problems when trying to implement such policies including distortions between public sector and private sector wages.

(5)

ii)

Taxation

Policy regarding taxation, its overall level and distribution between personal direct, indirect, corporate and other (e.g. stamp duty) will affect demand for goods and services, including labour, because of the impact on the prices. It will therefore affect the profitability of businesses and the (net) returns provided by investments.

Competition policy

Competition policy will be a crucial element of the operating environment, especially for the increasing number of naturally oligopolistic industries.

Labour policy

Labour policies set the background for the flexibility of labour and the bargaining power of organised labour. The cost of health services, welfare benefits and state pensions, are typically funded through taxation.

To the extent these are provided by government and paid for out of charges which are separate from taxation, they will have to be separately added to the cost of labour.

Government incentives for investment

Here we mean investment in the economic sense, i.e. the creation of new capital goods, e.g. a multinational company locating a car production plant, etc.

Incentives may take the form of grants, subsidies or tax breaks for specific types of investment or research and development.

Government funded, or otherwise encouraged, investment in infrastructure will help determine how easy it is for an organisation to put in place the physical requirements of its business within a territory.

(4)**[9 Marks]****Solution 5:****i)**

To improve future performance.

- If the performance is being measured, it might give the fund managers an extra incentive to maximise the returns of the funds they manage.
- finding out what has been successful in the past, investors should be better able to determine what might perform the best in the future, though with the understanding that past performance is not indicative of future returns.

Comparison of the rate achieved against a target rate.

- Many funds will have one or more target rate of return, e.g. the trustees of a pension fund will want to know how the rate of return achieved on the investments compared with:
 - the rate of return assumed in the actuarial valuation
 - the rate of salary increases for active members of the pension scheme.

Comparison against the performance of other funds, index or a benchmark portfolio.

- Those responsible for the funds will want to know how the performance of the portfolio compares with other similar portfolios.
- On the basis of this information, they will be able to make appropriate decisions regarding the future investment of the assets, e.g. whether a new fund manager should be hired, the asset allocation needs to be revisited, etc.

(5)

ii)

The difference in performance of two portfolios could be because the funds:

- have different investment strategies (eg one targets BSE100 and the other the Nifty50)
- are invested in different asset classes (eg one is a bond fund and one is an equity fund)
- are invested in different currencies
- have different investment restrictions or tracking error risk limits
- are of different sizes, which would lead to different spreading of fixed investment costs
- differ in style, for example, one may be actively managed and the other passively managed
- are subject to different management or investment fees
- have very different cashflows during the year, which can alter the money weighted performance
- have different tax statuses.

(4)

[9 Marks]

Solution 6:

i)

These three would be the sum of returns weighted by the exposure to respective asset classes for each.

- i. benchmark return = $0.70 \times 10\% + 0.2 \times 8.5\% + 0.1 \times 3.4\% = 9.04\%$
- ii. Manager A's actual portfolio return = $0.6 \times 12\% + 0.3 \times 9\% + 0.1 \times 2\% = 10.10\%$
- iii. Manager B's actual portfolio return = $0.40 \times 7\% + 0.4 \times 11.5\% + 0.2 \times 4.3\% = 8.26\%$

(3)

ii)

A strategic asset allocation (SAA) sets the long term investment strategy. This asset allocation is structured to meet the investment objectives of the fund.

Typically this would be prescribed as a target asset mix to the fund manager.

A tactical asset allocation (TAA) involves a short term deviation from the SAA with an intent to make additional investment returns. Typically TAA would be provided as a $\pm$ limit to the prescribed SAA limits.

TAA is guided by the belief that current investment markets are misaligned and they would correct shortly, after which the asset allocation would be expected to revert to SAA. Hence, it gives the fund manager freedom to leverage on his experience and apply expert judgment, in order to maximise returns for the fund, while staying within the permitted guidance.

(4)

iii)

The asset allocation profit and stock selection profit to be computed from the same table above.

- i. asset allocation selection profit is the TAA position taken by Manager A, given the benchmark portfolio returns.

$$= (0.6-0.7)*10\% + (0.3-0.2)*8.5\% + (0.1-0.1)*3.4\% = -0.15\%.$$
- ii. stock selection profit is the differential return made by investing in stocks different from the benchmark, given the asset allocation

$$= 0.4*(7\%-10\%) + 0.4*(11.5\%-8.2\%) + 0.2*(4.3\%-3.4\%) = +0.18\%$$

(2)

iv)

Manager A has not added value through TAA position (negative 0.15%), but has added value through the stock selection made within the asset classes (positive 1.21%).

Manager B has not added value through TAA position (negative 0.96%) but has added value through the stock selection made within the asset classes (positive 0.18%).

(2)

[11 Marks]

Solution 7:

- i) On the fixed side, interest of INR 12 million is earned at the end of the first year. This compounds to $12 \times 1.058 = \text{INR } 12.696$ million at the end of the second year.

A second interest amount of INR 12 million is added at the end of the second year, bringing the total compounded forward amount to INR 24.696 million.

This compounds to $24.696 \times 1.058 = \text{INR } 26.128$ million by the end of the third year, when there is the third interest amount of INR 12 million.

The cash flow at the end of the third year on the fixed side of the swap is therefore INR 38.128 million.

On the floating side we assume all future interest rates equal the corresponding forward MIBOR rates. Given the MIBOR zero curve, this means that we assume all future interest rates are 7% with annual compounding.

The interest calculated at the end of the first year is INR 14 million. Compounding this forward at 6.9% (forward MIBOR minus 10 basis points) gives $14 \times 1.069 = \text{INR } 14.966$ million at the end of the second year.

A second interest amount of INR 14 million is added at the end of the second year, the compounded forward amount is INR 28.966 million.

Compounding forward to the end of the third year, we get $28.966 \times 1.069 = \text{INR } 30.965$ million.

Adding in the final interest gives INR 44.965 million.

The value of the swap to Company Z (receiving floating)

$$= \frac{(44.965 - 38.128)}{1.07^3} = \text{INR } 5.58 \text{ million.}$$

(8)

- ii) Unless the portfolio to be hedged behaves exactly the same as the underlying index, the hedge will not be perfect.

For example, where a hedge ratio has been calculated to allow for the different volatility of a bond portfolio and the notional bond, there is a danger that the yield curve changes *shape* so the prices of bonds do not simply move in proportion to their volatility.

(2)

iii)

The standard deviation of ΔS is 0.0197.

The standard deviation of ΔF is 0.0207.

The coefficient of correlation between the two is 0.9219.

Hence, the optimal hedge ratio would be

$$= 0.9219 \times \frac{0.0197}{0.0207} = 0.8778$$

$$\text{Hence, the airline should take long position in} = 0.8778 \times \frac{1,000,000}{20,000} \\ = 43.89$$

$$= 44, \text{ rounding to the nearest whole number}$$

(7)

[17 Marks]

Solution 8:

i)

To protect or improve the strength and/or profitability of the dominant company, basically to maximize shareholder net worth.

Mergers sometimes happen because insurers may want to diversify across product offering and categories or get access to another distribution network or demographics segment or risk categories.

There could be possible tax advantages, such as a tax loss carry forward. If one of the firms involved in the merger has previously sustained net losses with no likely visibility of using these carry forward losses, these losses can be offset against the profits of the firm that it has merged with, a significant benefit to the newly merged entity.

Economies of scale is an often employed way of increasing operational efficiency. The cost of doing business generally decreases, e.g. acquisition costs may come down or the overheads may be spread over a bigger base, thereby leading to reduced unit costs.

One of the less admirable motives could be to protect a different merger that might be underway.

(3)

ii)

The intent of the model should be to estimate the market cap of the insurance business, which when divided by the share capital, would give the share price of this unlisted entity.

The market cap or company valuation of an insurance company comprises of two distinct parts – existing in-force book as on valuation date and the future new business.

The emergence of future profits, based on expected assumptions and projections, from the in-force book need to be estimated. The discounted value of these cashflows would be one component of value (DCF_1).

The insurance company would continue to write future new business, on a going concern basis. For a single block of new business written, emergence of future profits for that block on a standalone basis can be estimated based on expected assumptions and projections, in a manner similar to the one used for estimating the in-force book above. Hence, a standalone DCF model would give the value generated by one cohort of new business (DCF_{NB}).

The future new business would be expected to grow year on year (say at growth rate 'g'). Assuming same set of assumptions, product mix and business mix, DCF_{NB} could be used as a factor to project the value estimated in future and then discounted to give the second component of value (DCF_2), where r is the risk discount rate.

$$DCF_2 = DCF_{NB} \times (1+g) \times (1+r)^{-1} + DCF_{NB} \times (1+g)^2 \times (1+r)^{-2} + DCF_{NB} \times (1+g)^3 \times (1+r)^{-3} + \dots + DCF_{NB} \times (1+g)^n \times (1+r)^{-n}, \text{ with } n \text{ tending till perpetuity}$$

This can be estimated through a sum of an infinite geometric projection, i.e.

$$DCF_2 = DCF_{NB} / \{1-(1+g)/(1+r)\} = DCF_{NB} \times (1+r)/(r-g)$$

In case the assumptions vary from year to year resulting in different DCF_{NB} for each future year, a nested set of cashflows could be determined to estimate combined profit emerging in future by writing future new business after the valuation date.

Any adjustments towards tax or opportunity cost of holding regulatory solvency capital or any other encumbered capital requirement, could either be applied as a haircut on the aggregate number or estimated by an enhancement to the above DCF workings.

Sum of DCF_1 and DCF_2 , along with the balance sheet surplus on valuation date, will be the total value of the life insurance business.

(10)

iii)

From the working and formulae in the previous response, it is evident that the critical assumption is growth (g) – greater the growth assumption, greater is the value of future new business and hence the estimated market cap of the insurance company.

Further, because of the gearing effect, the gap between growth (g) and risk discount rate (r) would matter significantly. For example, if $r = 8\%$ and $g = 10\%$, an increase of an additional 2% in growth rate (which is only 20% greater new business) would have an effect - far in excess of 20% on the company value - because of the gearing effect.

(2)

[15 Marks]
